

SENTINELAI

Autonomous AI Security Operations Center

FINAL PROJECT • TECHNICAL + PRODUCT REPORT

Student: Fatima Bibi

Student ID: 977719

Defensive Multi-Agent Cybersecurity

1. Problem Statement

Security teams face large volumes of authentication, server, network and firewall events. Manual correlation and prioritization consume analyst time and can delay response.

2. Solution Overview

SentinelAI is a defensive multi-agent SOC prototype. It converts authorized security telemetry into a structured incident, enriches indicators, investigates evidence, scores risk, maps MITRE ATT&CK; recommends a response, requires human approval, simulates a controlled action and verifies the outcome.

3. What Is the Project?

A professional SOC command center that demonstrates agentic planning, specialist collaboration, persistent incident/audit state and safe response controls.

4. System Architecture

Security Sources → Ingestion/Normalization → SOC Orchestrator/Planner → Log Analysis → Threat Detection → Threat Intelligence → Investigation → Risk Assessment → MITRE ATT&CK; → Incident Response → Human Approval → Controlled Action → Verification → Audit Memory/Report.

5. Core Features

Covered inputs include server, network, authentication and firewall logs. The prototype supports SSH brute-force, network port-scan and repeated suspicious firewall-traffic scenarios, plus evidence timelines, risk/confidence, MITRE mapping, approval, verification, incident reports and audit memory.

6. Technologies Used

Python, Streamlit, SQLite, modular agent classes, regular-expression log parsing, structured agent results and MITRE ATT&CK; technique mapping.

7. Agentic Workflow

OBSERVE → REASON → PLAN → HUMAN APPROVAL → ACT → VERIFY. The orchestrator delegates to specialist agents and passes structured results rather than returning one chat answer.

8. Flagship Use Case

60 synthetic failed SSH logins from 185.77.21.44 target the admin account. SentinelAI detects brute force, enriches the IOC, correlates evidence, scores risk, maps T1110, prepares containment, waits for authorization and verifies a controlled simulation.

9. Demo

Executive Dashboard → Customer Demo or Live Detection → Agent Command Center → Investigation → Response Center → Approval → Controlled Containment → Verification → Incident Reports.

10. Target Audience

SMBs, universities, MSSPs, software/SaaS teams and internal IT/security teams.

11. Learning Outcomes

Multi-agent orchestration, security event parsing, threat detection, evidence correlation, risk scoring, explainability, human-in-the-loop safety, auditability and product design.

12. Challenges

Maintaining a transparent decision chain, keeping automated response safe, supporting multiple security-log types and persisting incidents/audit history.

13. Future Improvements

Authenticated Wazuh/Elastic/Splunk/Microsoft Sentinel connectors; real threat-intelligence APIs; ML anomaly detection; RBAC/SSO; cloud persistence; alerts; multi-tenancy; compliance reporting; and security-tested response playbooks.

14. Deployment

The package includes Render configuration. Local execution is recommended for the academic demo. Production requires secure secrets management, authentication, monitoring, persistence and security testing.

15. Product / Commercial Vision

Starter: local/synthetic telemetry. Pro: SIEM/TI integrations and alerting. Enterprise: RBAC/SSO, multi-tenancy, immutable audit, cloud deployment and secure response connectors. The academic version is a product foundation, not a claim of production certification.

16. Safety & Ethics

The prototype is defensive and uses synthetic/authorized data. It does not exploit, scan or modify real infrastructure. Consequential actions are explicitly human-gated and simulated.

Final Presentation Positioning

SentinelAI is designed to satisfy the Cybersecurity SOC Agent brief while presenting a polished product experience: visible specialist agents, explainable decisions, safe response, verification and persistent auditability.